

Project Design Phase

Problem – Solution Fit

Date	20 July 2026
Team ID	
Project Name	Plugging Into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Problem – Solution Fit

The problem: electricity consumption data across Indian states is scattered, inconsistent, and hard to interpret in raw form, leaving stakeholders unable to easily compare regions or spot trends.

The solution fit: a structured pipeline that cleans the data in Microsoft SQL Server, aggregates it with SQL queries, visualizes it through interactive Tableau dashboards, and presents it through a Flask web application — matching exactly how stakeholders want to consume the insights (visually, on demand, without needing SQL or Tableau Desktop access).

Existing Situation vs. Improved Situation

Aspect	Existing Situation	With the Proposed Solution
Data	Scattered, inconsistent raw files	Centralized, cleaned SQL Server tables
Analysis	Manual spread sheet comparisons	Structured SQL aggregation queries
Insight delivery	Static reports/spread sheets	Interactive Tableau dashboards
Access	Requires direct file/tool access	Accessible via a simple web application